Assignment 3: FSM Design for Traffic Light Controller Using GenAI

1. GenAI Model Name

ChatGPT (OpenAI)

2. Prompt

Design a Finite State Machine (FSM) for a traffic light controller with Green, Yellow, Red and Pedestrian signals. Use two D flip-flops to represent four states. The pedestrian request input should determine whether the controller enters the pedestrian state. Provide the state transition table, Boolean equations, and circuit implementation for verification in Logisim Evolution.

3. Response

The FSM uses two D flip-flops to represent four states:

- S0 (00): Green light ON.
- S1 (01): Yellow light ON.
- S2 (10): Red light ON.
- S3 (11): Red and Pedestrian signals ON.

The state sequence is S0 → S1 → S2. From S2, if the pedestrian request P = 1, the next state is S3; otherwise, it returns to S0. From S3, the controller returns to S0.

State transition table

Present State (Q1Q0)	P	Next State
00 (Green)	0 or 1	01
01 (Yellow)	0 or 1	10
10 (Red)	0	00
10 (Red)	1	11
11 (Pedestrian)	0 or 1	00

Next-state Boolean equations

$D1 = Q1\overline{Q}0$

$D0 = Q0\overline{(Q1}\overline{} + P)$

Output equations

$G = Q1\overline{Q}0\overline{}$

$Y = Q1\overline{Q0}$

R = Q1

Ped = Q1Q0

4. Verification by Logisim

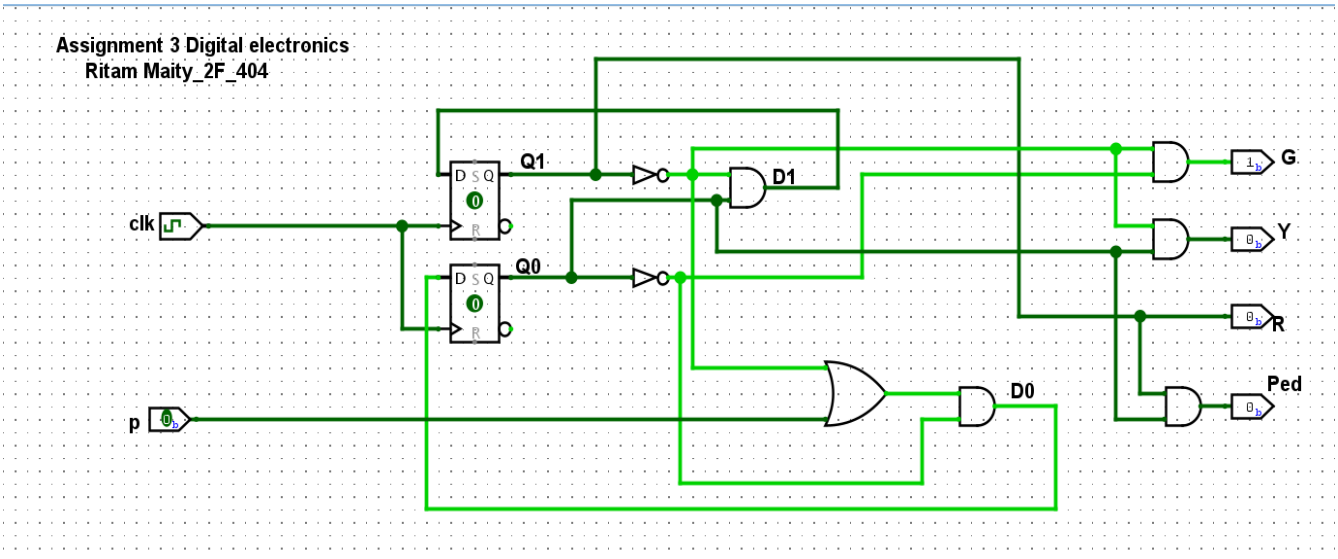
The circuit was constructed in Logisim Evolution using two D flip-flops, NOT gates, AND gates, an OR gate, a clock input, a pedestrian input and output indicators.

The circuit should be tested by resetting the simulation, applying clock pulses and observing the state outputs. The pedestrian input P is changed to test the transition from state 10 to state 11.

5. Circuit:

The circuit consists of three sections:

1. **Next-state logic:** NOT, AND and OR gates generate D1 and D0.
2. **State memory:** Two D flip-flops store Q1 and Q0 and receive a common clock.
3. **Output logic:** Gates generate Green, Yellow, Red and Pedestrian outputs.



6. Output

Expected output sequence when P = 0:

00 → 01 → 10 → 00

When P = 1 while the current state is 10:

10 → 11 → 00

State	Green	Yellow	Red	Pedestrian
00	1	0	0	0
01	0	1	0	0
10	0	0	1	0
11	0	0	1	1

7. Discussion

The FSM demonstrates how a traffic light controller can be designed using two D flip-flops and combinational logic gates. The pedestrian input controls whether the controller enters the pedestrian state. Logisim Evolution can be used to test the state transitions and output signals. This is a simplified model and does not include real-time traffic light delays.

8. GitHub Link:

https://github.com/Ritam-Maity/Digital-Electronics-Lab/tree/main

.